

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2169

Orchid 69 kV
March 27, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2169. The Interconnection Customer proposes a 84.9 MW solar photovoltaic facility interconnecting at the Orchid 69 kV substation in the North Carolina area. The request was received on August 29, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 580 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$491,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mccartney - North Tower Tap 345 kV L	Bear - Mecklenburg Power Sta 345 kV	106.3	1,698	14.4
N Towr Tap - Hagin 345 kV Line	Bear - Mecklenburg Power Station 345 kV	100.0	1,742	14.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mccartney - N Tower Tap Substation 345 kV	Reconductor line	\$33,872,000	1% (\$316,000)
N Towr Tap - Hagin 345 kV Line	Reconductor line	\$18,012,000	1% (\$175,000)
	Total allocated to project		\$491,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.